

Project Asteria - Robotics 101 Notes (Build Phase Intro)

Core Robot Concepts

- Robot pose = (x, y, theta) - theta = orientation (heading angle) - State defines everything needed to predict motion

Motors and Control

- DC motor provides rotation - PWM controls effective voltage and speed - H-Bridge controls direction - Inertia affects stopping time

Servo Motors and Feedback

- Servo = motor + gearbox + sensor + controller - Closed-loop system - Error = target - current position - Continuous correction based on feedback

Encoders and Odometry

- Encoders measure wheel rotation - Distance = revolutions $\times$ wheel circumference - Update equations: $x \pm d \cos(\theta)$ $y \pm d \sin(\theta)$ - Enables self-localization

Differential Drive

- Two wheels control motion - Same speed $\rightarrow$ straight line - Different speeds $\rightarrow$ curved path - Opposite speeds $\rightarrow$ rotation in place

Control Systems

- Feedback compares desired vs actual state - Overshoot due to inertia - PID: P = proportional I = integral D = derivative (predictive damping)

Project Asteria Design

Structure: - robot.py (state) - physics.py (motion equations) - controller.py (decisions) - world.py (environment) - main.py (simulation loop) Goal: Differential Drive Robot Simulator

Engineering Mindset

- Model before coding - Separate control and action layers - Build iteratively - Errors accumulate in real systems